Early Number Systems — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1	**(c) 50.** L = 50.
Q2	**(b) XLIX.** 49 = 40 (XL) + 9 (IX).
Q3	**715.** 500 + 100 + 100 + 10 + 5.
Q4	**(c) IL.** The valid pairs are IV, IX, XL, XC, CD, CM — not IL.
Q5	**True.** Roman numerals are additive with no place value, so X is always 10.
Q6	**(a) Both true, R explains A.** Since IIII is not used, 4 is written as IV.

Short answer

Q7	**(i) MCCXXII (ii) CCCII.**
Q8	**(i) 2999 (ii) 715.**
Q9	**MMCCCLXVII.** 2367 = 2000 + 300 + 60 + 7 = MM + CCC + LX + VII.

Long answer

Q10	**(i) CDXLIV (ii) MCMLXXXIV.** 444 = CD + XL + IV; 1984 = M + CM + LXXX + IV.
Q11	**Any three:** it has no zero; it has no place value; it needs many symbols for large numbers; and multiplication/division are very hard to perform.
Q12	**21 symbols.** 70707 = 7 ten-thousands + 0 thousands + 7 hundreds + 0 tens + 7 ones = 7 + 7 + 7 = 21 symbols.

Total: 21 marks.